Project Title

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https://github.com/hourlilies/elec5305-project-550790019

August 26, 2026

1 Project Overview

This project aims to develop a synthesizer, sampler, and music sequencer for any device capable of running a modern web browser supporting WebAudio, using the emscripten compiler toolchain that turns C++ DSP code into WebAssembly. The goal is to make music production more accessible, because pretty much any computer, whether that be a smartphone, laptop, or even a single-board computer, can run a modern web browser that supports WebAudio. This is important because anyone should be able to play with and make music, without needing to install and learn an OS-dependent digital audio workstation, packed full of comprehensive features but designed for professional musicians and not them.

Though the advent of generative AI has lowered the barrier of entry to music creation tremendously, empowering authors to create music to their taste with just a few text prompts and a bit of luck, it has come at the cost of abstracting away the underlying signal processing and audio theory that makes it all work. This project aims to expose all of that, the wires, knobs, and switches of music making, with the hopes of allowing the prospective musician to see and poke at the machine that makes modern music tick, whose invention and decades of production is what allows powerful GenAI tools like Suno or Udio to even exist.

2 Background and Motivation

3 Proposed Methodology

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4 Expected Outcomes

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5 Timeline

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6 References

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7 Appendix

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